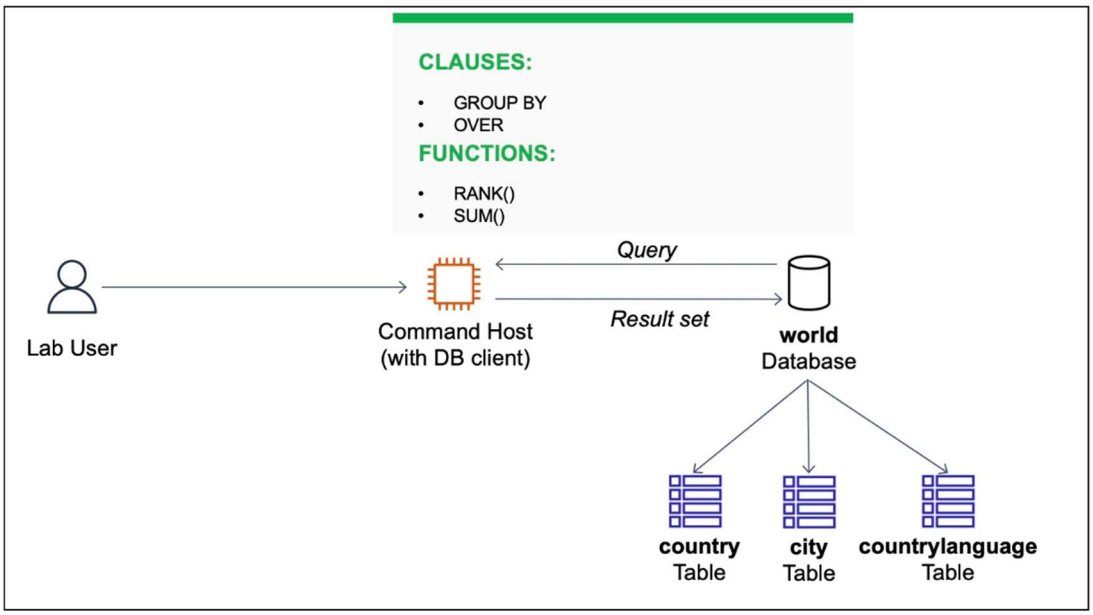
273--Lab - Organización de datos

En esta práctica se muestra el funcionamiento de los siguientes comandos en SQL:

- GROUP BY para agrupar los datos
- OVER() para realizar cálculos por grupo sin necesidad de agrupar visualmente
- RANK() para asignar números de clasificación

Escenario

Se cuenta con una instancia EC2 configurada con un cliente de base de datos MySQL.



Para conectarnos a la instancia y a la base de datos seguimos los pasos de las prácticas anteriores.

Hacer consultas a la base de datos

Verificamos las columnas y datos de la tabla *country*

```
MariaDB [(none)]> SELECT * FROM world.country;
+-------+-----------------------------------------------+---------------+---------------------------+-------------+
-------+-----------------------------------------------+-----------------------------------------------+---------+-
| Code  | Name                                          | Continent     | Region                    | SurfaceArea |
ld      | LocalName                                     | GovernmentForm                                | Capital |
+-------+-----------------------------------------------+---------------+---------------------------+-------------+
-------+-----------------------------------------------+-----------------------------------------------+---------+-
| ABW   | Aruba                                         | North America | Caribbean                 |      193.00 |
793.00 | Aruba                                         | Nonmetropolitan Territory of The Netherlands |     129 |
| AFG   | Afghanistan                                   | Asia          | Southern and Central Asia |   652090.00 |
  NULL | Afganistan/Afqanestan                         | Islamic Emirate                               |       1 |
| AGO   | Angola                                        | Africa        | Central Africa            |  1246700.00 |
984.00 | Angola                                        | Republic                                      |      56 |
| AIA   | Anguilla                                      | North America | Caribbean                 |       96.00 |
  NULL | Anguilla                                      | Dependent Territory of the UK                 |      62 |
| ALB   | Albania                                       | Europe        | Southern Europe           |    28748.00 |
500.00 | Shqipëria                                     | Republic                                      |      34 |
| AND   | Andorra                                       | Europe        | Southern Europe           |      468.00 |
```

Filtramos por los países de la región de Australia and New Zealand y ordenamos por la población en orden descendente

```
MariaDB [(none)]> SELECT Region, Name, Population FROM world.country WHERE Region = 'Australia and New Zealand' ORDER By Population desc;
+---------------------------+-------------------------+------------+
| Region                    | Name                    | Population |
+---------------------------+-------------------------+------------+
| Australia and New Zealand | Australia               |   18886000 |
| Australia and New Zealand | New Zealand             |    3862000 |
| Australia and New Zealand | Christmas Island        |       2500 |
| Australia and New Zealand | Norfolk Island          |       2000 |
| Australia and New Zealand | Cocos (Keeling) Islands |        600 |
+---------------------------+-------------------------+------------+
5 rows in set (0.000 sec)
```

Con GROUP BY agrupamos la consulta anterior por región y con SUM() obtenemos la población total de la región

```
MariaDB [(none)]> SELECT Region, SUM(Population) FROM world.country WHERE Region = 'Australia and New Zealand' GROUP By Region ORDER By SUM(Population) desc;
+---------------------------+-----------------+
| Region                    | SUM(Population) |
+---------------------------+-----------------+
| Australia and New Zealand |        22753100 |
+---------------------------+-----------------+
1 row in set (0.002 sec)
```

La cláusula OVER()se utiliza para definir un conjunto de filas (una "ventana") sobre las cuales se realiza un cálculo, sin necesidad de agrupar los datos en una sola fila como sucede con GROUP BY.

Usando OVER() agrupamos por región y con SUM() obtenemos la población total de la región.

```
MariaDB [(none)]> SELECT Region, Name, Population, SUM(Population) OVER(partition by Region ORDER BY Population) as 'Running Total' FROM world.country WHERE Region = 'Austra
lia and New Zealand';
+---------------------------+-------------------------+------------+---------------+
| Region                    | Name                    | Population | Running Total |
+---------------------------+-------------------------+------------+---------------+
| Australia and New Zealand | Cocos (Keeling) Islands |        600 |           600 |
| Australia and New Zealand | Norfolk Island          |       2000 |          2600 |
| Australia and New Zealand | Christmas Island        |       2500 |          5100 |
| Australia and New Zealand | New Zealand             |    3862000 |       3867100 |
| Australia and New Zealand | Australia               |   18886000 |      22753100 |
+---------------------------+-------------------------+------------+---------------+
5 rows in set (0.002 sec)
```

Diferencia visual entre GROUP BY y OVER():

- GROUP BY: Si tenemos10 filas de ventas y agrupamos por "País", obtendremos una sola fila por país con el total.

- OVER(): Si suamos SUM(monto) OVER(PARTITION BY pais), seguiremos viendo las 10 filas individuales, pero al lado de cada venta veremos el total de ese país.

La función **RANK()** es una **función de ventana** (window function) que se utiliza para asignar un número de clasificación o "ranking" a cada fila dentro de una partición de un conjunto de resultados.

Usando RANK() asignamos un número de clasificación a cada país de la región de Australia and New Zealand con base a la población en orden ascendente

```
MariaDB [(none)]> SELECT Region, Name, Population, SUM(Population) OVER(partition by Region ORDER BY Population) as 'Running Total', RANK() over(partition by region ORDER BY
 population) as 'Ranked' FROM world.country WHERE region = 'Australia and New Zealand';
+-------------------------+-----------------------+------------+---------------+--------+
| Region                  | Name                  | Population | Running Total | Ranked |
+-------------------------+-----------------------+------------+---------------+--------+
| Australia and New Zealand | Cocos (Keeling) Islands |        600 |           600 |      1 |
| Australia and New Zealand | Norfolk Island          |       2000 |          2600 |      2 |
| Australia and New Zealand | Christmas Island        |       2500 |          5100 |      3 |
| Australia and New Zealand | New Zealand             |    3862000 |       3867100 |      4 |
| Australia and New Zealand | Australia               |   18886000 |      22753100 |      5 |
+-------------------------+-----------------------+------------+---------------+--------+
5 rows in set (0.002 sec)
```

Ahora asignamos un número de clasificación a cada país dentro de su región, con base a la población en orden descendente

```
MariaDB [(none)]> SELECT Region, Name, Population, RANK() over(partition by region ORDER BY population DESC) as 'Ranked' FROM world.country;
+-------------------------+---------------------------------------------+------------+--------+
| Region                  | Name                                        | Population | Ranked |
+-------------------------+---------------------------------------------+------------+--------+
| Antarctica              | French Southern territories                 |          0 |      1 |
| Antarctica              | Bouvet Island                               |          0 |      1 |
| Antarctica              | South Georgia and the South Sandwich Islands |          0 |      1 |
| Antarctica              | Antarctica                                  |          0 |      1 |
| Antarctica              | Heard Island and McDonald Islands           |          0 |      1 |
| Australia and New Zealand | Australia                                   |   18886000 |      1 |
| Australia and New Zealand | New Zealand                                 |    3862000 |      2 |
| Australia and New Zealand | Christmas Island                            |       2500 |      3 |
| Australia and New Zealand | Norfolk Island                              |       2000 |      4 |
| Australia and New Zealand | Cocos (Keeling) Islands                     |        600 |      5 |
| Baltic Countries        | Lithuania                                   |    3698500 |      1 |
| Baltic Countries        | Latvia                                      |    2424200 |      2 |
| Baltic Countries        | Estonia                                     |    1439200 |      3 |
| British Islands         | United Kingdom                              |   59623400 |      1 |
| British Islands         | Ireland                                     |    3775100 |      2 |
```